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Sensorimotor dysfunction following anterior cruciate ligament injury (Part 1). What is it? How can clinicians assess it?

Authors

Thilina N. Vitharana, PT, BSc, MSc [1, 2], Enda King, PT, PhD [3, 4], Neil Welch, PhD [1], Brian Devitt, MD, PhD [1,2], Kieran Moran, PhD [2, 5]

[1] Sports Medicine Department, UPMC Sports Surgery Clinic, Dublin, Ireland

[2] School of Health and Human Performance, Dublin City University, Dublin, Ireland

[3] Aspetar, Orthopaedic and Sports Medicine Hospital, FIFA Medical Centre of Excellence, Doha, Qatar

[4] Department of Life Sciences, Roehampton University, London, UK

[5] Insight Centre for Data Analytics, Dublin City University, Dublin, Ireland

Corresponding author: Thilina N Vitharana PT, UPMC Sports Surgery Clinic, Santry, Dublin 9, Ireland (email: vitharanat@upmc.ie) @Thilina_Physio

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Abstract

Background

The anterior cruciate ligament (ACL) plays an important sensory role within the sensorimotor system. Following ACL injury, sensorimotor dysfunction can have implications for rehabilitation and risk of reinjury.

Clinical question

What dysfunction occurs within the sensorimotor system following ACL injury, and how can clinicians assess it?

Key results

Following ACL injury, dysfunction occurs across the sensorimotor system: afferent pathways, efferent pathways and central processing. The afferent pathways exhibit dysfunction within the somatosensory system ([1] increased pain and swelling, [2] increased central cortical processing and [3] reduced proprioception). There is also dysfunction in the visual system (increased visual-motor reliance and central cortical processing). The efferent pathways have reduced excitability of the central cortex, reduced descending motor pathway excitability and altered spinal reflexive excitability (acutely reduced but then chronically increased).

Clinical application

Protocols to assess athletes' sensorimotor function following ACL injury might help clinicians quantify the risk of reinjury. Assessing central processing requires specialised equipment, not typically accessible to clinicians. A practical approach to quantify the extent of sensorimotor dysfunction could focus on assessing the afferent and efferent pathways: tests of proprioception (e.g. joint position sense test), pain (e.g. visual analogue scale and numerical pain rating scale), swelling (e.g. sweep test and ballottement test), visual-motor reliance (e.g. stepdown test), visual-motor processing ability (e.g.

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sensory stations or neurocognitive tests), muscle strength (e.g. repetition maximum testing or isokinetic dynamometry) and voluntary activation (e.g. electromyography).

Key words

Anterior cruciate ligament, sensorimotor dysfunction, assessment, rehabilitation, afferent, efferent

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Introduction

The sensorimotor system is complex, and a central element of the body's motor control system that assists with planning and executing movement.[82] It encompasses the afferent sensory systems, the efferent motor pathways and the processing within the central cortex.[82] The afferent sensory systems include: visual, somatosensory (interpreting sensations from the skin and within the body such as pain, touch and vibration), vestibular (balance and spatial orientation), olfactory, gustation and auditory systems.[55,76] Integrating sensory information from afferent systems, especially the somatosensory and visual systems, allows the individual to gain accurate information from the environment to guide an appropriate movement strategy (Figure 1). The efferent motor pathways then signal the muscles to execute the desired movement strategy.

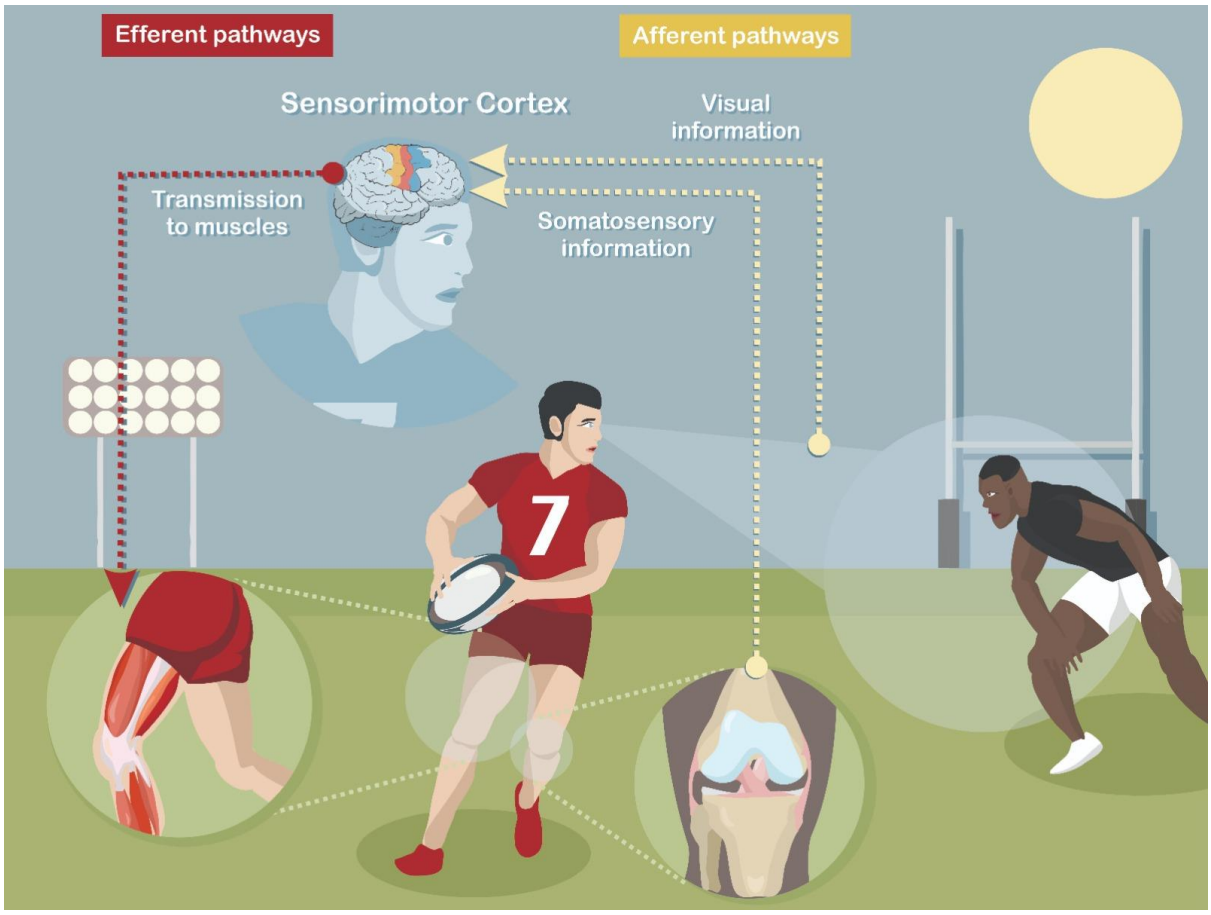

Figure 1. An individual uses somatosensory and visual systems to gather information that helps the sensorimotor cortex create an appropriate motor response.

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Musculoskeletal injuries, including anterior cruciate ligament (ACL) injuries, lead to sensorimotor system changes, and ultimately sensorimotor dysfunction.[27,33,40,57,61] The dysfunction manifests as errors in planning and executing movement.[94] Sensorimotor dysfunction following ACL reconstruction (ACLR) may be a risk factor in sustaining a new ACL injury,[28,94] even after completing rehabilitation. Therefore, clinicians must understand how sensorimotor dysfunction occurs following ACL injury, how to assess it and rehabilitate it.

This clinical commentary outlines sensorimotor changes following an ACL injury, and proposes evidence-based methods to identify dysfunction, that are driven by our clinical practice. In part 2, we focus on rehabilitation techniques for sensorimotor dysfunction.

Clinical Question

What sensorimotor dysfunction occurs following ACL injury and how can clinicians assess it? The ACL has two important roles: (1) providing structural knee stability by limiting excessive anterior translation and internal rotation of the tibia, and knee valgus,[11,89] and (2) providing sensory information to the sensorimotor cortex to facilitate motor planning and execution.[22,68] Even when structural stability is restored after ACLR, sensorimotor dysfunction remains,[24,27] which may impair gathering sensorimotor information from the environment when planning movement. Alterations in efferent motor pathways remain following ACLR, even up to two years post-surgery with reduced strength and activation of the quadriceps.[31,35]

Clinicians may lack either the clinical tools to identify individuals with sensorimotor dysfunction, or rehabilitation methods may be inadequate. Dysfunction in the sensorimotor system following ACLR is the same as for ACL injuries that are managed non-operatively and therefore, the two conditions (ACLR and ACL injuries) are not separated.

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How does sensorimotor dysfunction impact movements?

Motor planning and execution is complex. Neurocognitive demands (processing sensory, motor, emotional and higher executive information) are increased in sporting situations due to movement speed, greater decision-making complexity and reduced decision-making time. An athlete's movement strategy is determined by their constraints (anthropometrics and neuromuscular capacities), the perceived environment information (opposition position and movement) and task constraints (e.g. attempting to score a goal).[2,12,102] Perception of environmental information and action continuously occurs,[20] requiring the athlete to integrate sensory information to generate an appropriate movement strategy. Individuals have several sensory systems: visual, somatosensory (interpreting sensations from the skin and within the body such as pain, touch and vibration), vestibular (balance and spatial orientation), olfactory (smell), gustation (taste) and auditory (hearing),[55,76] with the somatosensory and visual systems being the most pertinent for motor planning and execution. Therefore, sensorimotor dysfunction can increase neurocognitive demands or reduce the ability to plan high impact movements efficiently (such as cutting or jumping/landing).

Key results

Somatosensory system

The somatosensory system (1) informs the central cortex about the position and movement of body segments (proprioception) and, (2) senses painful and pressure/vibration stimuli.[76] Proprioception uses fast velocity afferent fibres (within muscle, ligament and joint tissue) to provide starting posture information so that appropriate coordination of motor units can be produced to execute a movement.[75] The second somatosensory function uses soft tissue pain and pressure receptors to detect pain or swelling so that if an injury occurs or tissue damage is imminent, then the body can coordinate a motor response to minimise damage or protect the injured structure.[81,86]

A full-thickness ACL tear results in a complete loss of mechanoreceptors.[14] The return of these mechanoreceptors following ACLR is often slow and variable,[14] which may lead to altered somatosensory processing and a reorganisation of the sensorimotor cortex with greater reliance being

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placed on the visual system.[57] Diminished proprioceptive ability following ACLR may mean greater error for the athlete when determining how their body is positioned in the environment. Knee proprioception is reduced after ACLR, compared to both the uninjured limb and healthy control limbs, even after rehabilitation and return to sport.[42,79,80] Changes in surgical technique to preserve the remnants of the ACL during reconstructive surgery may improve proprioception,[7] but the surgical technique is uncommon.

Pain and swelling following an ACL injury may inhibit quadriceps activation.[81] Knee joint effusion causes activation of afferent pressure receptors within the joint capsule[60] due to high intra-articular pressures, and chemical activation of the receptors due to inflammatory cytokines.[81] Afferent pressure receptor activations may reduce quadriceps activity[64] and strength. [53] Similarly, pain and inflammation can sensitise afferent receptors within the joint capsule and connective tissues, leading to further quadriceps inhibition,[60,81] with strength losses of 80-90% within the first few days following knee surgery, which may persist in the weeks after.[87] It is therefore essential to measure and monitor athlete pain and effusion throughout the rehabilitation process.

Assessment

Assessing somatosensory function involves assessing somatosensory cortical processing demands, assessing proprioception, and assessing pain and effusion (Table 1). However, the former generally uses equipment (e.g. functional brain MRI) not readily accessible. Therefore, clinicians should focus on assessing proprioception, pain and effusion.

Proprioception

Examining proprioceptive function is difficult because it covers several domains: (1) joint position sense, (2) sense of movement (kinaesthesia), (3) sense of movement velocity, and (4) sense of force.[75] Methods to examine kinaesthesia, sense of velocity and sense of force use specialised tools, such as isokinetic dynamometers. However, joint position sense (JPS) testing is more easily accessible and demonstrates significant proprioceptive impairment in the ACLR limb[21,80] and we recommend its use.

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TABLE 1. Tests of somatosensory function

Somatosensory domain	Clinical test	Use	Equipment needed	Facilitators	Barriers
Somatosensory cortical processing	Functional brain MRI	Allows clinicians to determine if altered processing is occurring in somatosensory cortical areas during a movement task (eg. Knee flexion and extension)	MRI machine and specialist software	Able to identify areas of brain active during movement	Financial and time cost. Expertise to interpret the scan.
	Electroencephalogram	Allows clinicians to determine if altered processing is occurring in somatosensory cortical areas during a movement task (eg. Knee flexion and extension)	EEG device and specialist software	Able to identify areas of brain active during movement	Financial and time cost. Expertise to interpret the scan.
Proprioception	Joint position sense tests	Whilst their vision is obstructed, athletes have their knee moved to an angle by the assessor and the athlete is then asked to move the knee to replicate the angle	Isokinetic dynamometer	Reliable measurement	Cost of equipment
			Camera and angle calculator software	Easy to set up and reliable measurement	Can take time to setup, process images and measure angles
			Digital goniometer	Easy to set up and reliable measurement	
	Threshold to detect passive motion	Whilst their vision is obstructed, the athlete is instructed to report when they feel their knee move.	Isokinetic dynamometer	Reliable measurement	High cost

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	Ability to detect change in velocity	Whilst their vision is obstructed, the athlete reports to the assessor when they feel the speed of their knee movement changes	Isokinetic dynamometer	Reliable measurement	High cost
	Force reproduction tests	The athlete completes an isometric contraction in an attempt to hit a target threshold (generally 50% of maximum voluntary contraction).	Isokinetic dynamometer	Reliable measurement	High cost
Pain	Visual Analogue Scale (VAS)	The athlete rates the magnitude of their knee pain by making a mark on a 10cm line (0= no pain and 10= worst pain imaginable)	Paper and ruler	Quick and reliable measurement	The assessment is very subjective so can vary due to non-physical factors
	Numerical Pain Rating Scale (NPRS)	The athlete rates the magnitude of their knee pain from 0 to 10 (0= no pain and 10= worst pain imaginable)	No equipment	Quick and reliable measurement	The assessment is very subjective so can vary due to non-physical factors
Effusion	Sweep test	Assessor uses sweeping motion around the knee to measure the effusion.	No equipment	Quick and simple test	Inter-observer reliability can be varied
	Patellar Ballottement test	Assessor applies pressure to the knee and assesses the degree to which the patella floats to determine level of effusion	No equipment	Quick and simple test	Inter-observer reliability can be varied

JPS testing can be split into three broad categories: image/photograph recorded angulation testing, electrogoniometry, and isokinetic dynamometry. While all three approaches can be reliable,[19,59,74,78] some protocols have poor reliability.[90] Larger absolute errors between target angle and athlete's angle replication attempt indicate JPS deficit. An image-recorded angulation JPS testing method can be easily implemented in the clinic and with good reliability (ICC=0.96-0.98) (Appendix A).[77-79] Among 10 elites athletes with ACLR, there was a difference of 5.3° in absolute error between the ACLR limb (absolute error= 7.2°) and the uninjured limb (absolute error= 1.9°), reflecting ACLR proprioceptive dysfunction.[79]. We recommend JPS testing every four to six weeks during rehabilitation.

Assessing pain and effusion

Regularly assessing joint pain and effusion is essential to minimise potential muscle inhibition, particularly in the quadriceps and hamstrings,[54] and to ensure that the athlete's knee is tolerating rehabilitation program progressions. Numeric rating scale and visual analogue scale are valid and reliable measures, so select a method based on what the athlete prefers.[99] A minimal clinically important change in numeric rating scale has not been examined in the ACLR population; it is 2 points in people with chronic musculoskeletal pain.[85] The visual analogue scale requires athletes to mark their pain on a 10-centimetre line between '0' (no pain) and '10 centimetres' (worst pain imaginable). A minimal clinically important change of 1.4 centimetres is suggested for musculoskeletal conditions and chronic pain.[13,39] If pain persists, then consider strategies to manage pain, especially during early stage rehabilitation as pain may inhibit muscle recruitment[34] and result in limb off-loading strategies. During mid-to-late stage rehabilitation, pain that negatively affects execution of sporting motor tasks or rehabilitation exercises also needs addressing. Pain assessments should be completed every session with the clinician until the pain resolves to assess how the athlete is responding to their interventions.

Measuring effusion following ACL injury is not a direct measure of afferent function but allows clinicians to assess a secondary factor (pressure or inflammation) that may be inhibiting mechanoreceptors within and surrounding the knee. The sweep test and the ballottement test (Appendix B) both have good inter-observer agreement[8]. Effusion testing should be completed every

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session the athlete has with the clinician until resolved to ensure that the knee is responding appropriately.

Visual system

The visual system may be the predominant sensory system used when planning and executing motor tasks.[16,23,93] When there is conflict between somatosensory system and visual system afferent information, the visual information takes precedence.[70,93] The common thread through theories of how visual information is processed is that the information from the visual system provides the central cortex with information regarding what is in the environment and where it is positioned.[16]

There is limited research examining the changes that occur to the visual system and how visual information is processed following ACL injury. The term visual-motor system encompasses all visual processes and how they influence motor actions.[4,23,24,26] Functional brain MRI studies show increased activity in the visual processing cortices following ACLR.[27,61] Increased visual processing activity may stem from loss of proprioception during ACL injury and the subsequent effusion and nociception.[24] Altered afferent information to the sensorimotor cortex leads to reduced postural control, and increased demands on motor planning centres in the frontal lobe and secondary somatosensory cortices (as evidenced by functional brain MRI).[6,27,57] To manage reduced postural control and knee afferent input, a re-weighting of the sensory systems may occur with an increased reliance placed on visual information during motor planning and execution.[27] This re-weighting potentially contributes to increased cognitive demand which may be detrimental to an athlete by reducing their ability to react quickly to their changing environment.[6,37] Given the potential link between increased reliance on the visual-motor system and ACLR reinjury,[28] it is important that clinicians assess for potential increased visual reliance.[25]

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Assessment

Assessment of visual function commonly covers two domains: visual-motor processing ability and increased reliance on the visual-motor system via balance tests (Table 2).

Visual-motor processing ability

This encompasses all processes that occur from an athlete using their eyes to capture environmental information, the processing of the information in the central cortex and its influence on motor execution.[96,97] Poor visual-motor processing ability (visual memory, reaction time and processing speed) may contribute to non-contact ACL injuries as assessed with the ImPACT computer-based neurocognitive test (ImPACT Applications Inc.).[95]

Different companies (e.g. ImPACT [https://impacttest.com/], and Cogstate [https://www.cogstate.com/]) provide computer-based neurocognitive tests that are easy for clinicians to administer at a relatively low cost, even on a large scale. These tests are reliable in uninjured athletes[58] but no study has examined their reliability in the ACLR population. A common issue with the neurocognitive tests is that they predominantly test memory and speed of processing domains, with a lack of varied visual-motor domains.

'Sensory stations' and tablet systems (e.g. Senaptec, Oregon USA) with visual-motor specific testing make it easier to assess across a number of visual-motor domains that reflect visual acuity (discern shapes and object details), visual memory (store visual information so that learning can occur from previous experiences), near-far eye movements (switch gaze from near to far objects) and multiple object tracking ability (pick up multiple environmental cues) (https://www.youtube.com/watch?v=4Vj78RX7oVo).

The sensory station has good reliability in uninjured athletes,[17] but no studies have examined its reliability in the ACLR population . While measures of visual-motor performance can predict in-game performances in sports such as professional baseball,[41,49] there is currently no research examining ACL injury risk or changes in visual-motor function following injury . As such, there is limited information on what a good baseline score is for each of the visual-motor domains. For the tests to

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be beneficial, athlete baseline information is required to determine if function has been negatively affected by injury. We have found athletes enjoy engaging in the sensory station due to the novelty and the varied challenging tasks. Their results can be exported as raw scores so that deeper analysis can be conducted. The athlete is provided with an immediate percentile score (against other athletes from similar sports) which drives competition and engagement.

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TABLE 2. Tests of visual-motor function

Visual assessments	Clinical test	Use	Equipment needed	Facilitators	Barriers
Visual cortical activity	Functional brain MRI	Assess if altered processing is occurring in visual cortical areas during a movement task (eg. Knee flexion and extension)	MRI machine and specialist software	Able to identify areas of brain active during movement	Financial and time cost. Expertise to interpret the scan.
	Electroencephalogram	Assess if altered processing is occurring in visual cortical areas during a movement task (eg. Knee flexion and extension)	EEG device and specialist software	Able to identify areas of brain active during movement	Financial and time cost Expertise to interpret the scan
Visual-motor processing	Neurocognitive tests (e.g. ImPACT, Cogstate)	Assess domains such as: visual processing speed, reaction time and visual memory	Computer based test	Standardised testing procedure	Each testing session has a cost. Assesses a limited number of visual-motor processing domains.
	Sensory stations (e.g. Senaptec)	Assesses domains such as: visual acuity, depth perception, near-far eye movement, contrast sensitivity, visual memory and multiple object tracking	Computer/tablet based test	Is easy to administer to patients. Assesses across visual-motor processing abilities that may be relevant to athletes	Cost of device can be high. Needs further research in ACLR population
Visual-motor reliance	Single leg balance. Eyes open versus eyes closed	Assess postural control differences between eyes open and eyes closed conditions during a single leg balance task	Force plate is optional	Can use time holding the position or counting errors within specific time	Does not appear to be a good discriminator between ACLR

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				period as measure of postural control. Can use centre of pressure data if there is access to force plates.	and uninjured individuals in determining visual-motor reliance. If utilising force plates, then there is a cost.
	Stepdown test. Eyes open versus eyes closed	Measure time to stability during a stepdown task while eyes are open and eyes are closed. Clinicians can derive an index score of how visually-motor reliant an athlete may be	Force plate and step	Quick test and safe test to complete	Requires force plates to complete the test.

Assessing visual-motor reliance via balance tests

Balance tests are common in clinical settings to assess an athlete's ability to integrate sensory (somatosensory, visual and vestibular) information. The athlete performs a task that compares their balance under two conditions: eyes open versus eyes closed. A greater deficit in performance, as compared to either normative data or the uninjured limb, indicates an increased visual-motor system reliance.

Postural control worsens when vision is obstructed during single-limb standing in ACL deficient individuals as compared to healthy participants.[103] However, the ACLR population exhibits similar reductions in postural control performance as healthy participants (ACLR 43% reduction vs controls 45% reduction),[103] suggesting that either: (1) visual-motor reliance does not negatively affect an ACLR athlete's ability to maintain postural control in comparison to healthy controls, or (2) the task of quasi-static standing on one leg may not be challenging enough.

We adapted a 20cm stepdown task (Colby et al)[9] in which athletes step onto a force plate where they must attain stability as quickly as possible (Appendix C). Athletes complete this with eyes open and then eyes closed, with time to stability used as the outcome measure.[5,9] We found good reliability (ICCs= 0.71-0.96) (unpublished) and a visual-motor reliance index score can be ascribed to athletes. Our findings show that healthy athletes improved their time to stability time by 17% with eyes closed. Therefore, if an athlete's time to stability worsens with eyes closed, we attribute this to visual reliance and suggest they would benefit from targeted rehabilitation strategies. If clinicians do not have access to force plates, subjective assessment of postural control (assessing frontal plane control, trunk sway, hip movement, knee adduction, and foot pronation and supination) may be used to assess athlete performance. We reassess visual-motor reliance every six weeks.

Efferent system

The main efferent system components involved in motor planning and execution are the motor cortex, descending motor pathway and spinal-reflexive pathways.[73,104] After ACL injury, there are changes in all three components,[71] which are still evident 6 months following ACLR,[44] manifesting in changes in

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muscle activation, coordination[56] and reduced muscle strength.[43] This likely contributes (along with muscle atrophy) to persistent quadriceps weakness[65,73] and ongoing dysfunction.[50,72] Following ACLR, unilateral and bilateral deficits in quadriceps voluntary activation are present up to four years post-surgery.[48,71] Deficits in quadriceps activation can result in reduced strength and performance,[51] and increased risk of reinjury.[69] Although there is some debate,[71] the driver of the strength deficit is likely: (1) alterations to the efferent system (motor cortex, descending motor pathway and spinal-reflexive pathways), (2) local muscle atrophy, and (3) negative influences from changes in the sensory system (as previously discussed).

Motor cortex

The motor cortex (e.g. premotor cortex and primary motor cortex) is involved in movement planning and initiation.[47] Following ACLR, there is increased intracortical inhibition associated with reduced voluntary quadriceps activation in the injured limb,[52] which can occur bilaterally.[105] Subsequently, the athlete needs to dedicate greater neural resources and effort to activate a muscle and execute movement, compared to preinjury, thereby potentially resulting in non-optimal movement strategies.[6,25]

Descending motor pathway

The descending motor pathway (corticospinal tract) carries impulses from the motor cortex through the spinal cord to synapse with the lower motor neurone that activates the target muscle.[38,84] Its excitability is reduced following ACLR via an increased firing threshold[44,71] which may occur in response to increased motor cortex inhibition.[29,44] The increased threshold results in reduced quadriceps activation,[71] which may change movement strategies.

Spinal Reflexive pathways and peripheral changes

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Spinal reflexive pathways facilitate fast motor responses to peripheral stimuli. ACL injury and ACLR lead to pain, knee joint effusion and loss of proprioceptors within the native ACL,[66] which inhibits the quadriceps and increases hamstrings motor activity..[36,88] Over time, the excitability of the spinal reflexive pathway to the quadriceps increases such that it is significantly greater in both the ACLR limb and unreconstructed limb in comparison to healthy athletes.[83,98]

All components of the efferent system (motor cortex, descending motor pathway and spinal reflexive pathways) are influenced by ACL injury. An initial reduction in excitability occurs within the spinal reflexive pathways and motor cortex. Reductions in descending motor pathway volume and excitability can then develop, with changes occurring 3-6 months post ACLR.[45] These factors may culminate in reduced voluntary activation and function of the quadriceps,[71] negatively impacting strength and knee control improvements during rehabilitation.

Assessment

Assessing the efferent subsystems is primarily conducted with specialised testing equipment and expertise generally unavailable to clinicians. These include Hoffman reflex to assess spinal reflexive changes, transcranial magnetic stimulation to determine corticospinal excitability, and central activation ratio to measure voluntary activation levels of muscles.[44] Therefore, clinicians should measure muscle strength and activation (especially the quadriceps) to determine efferent function following ACLR. These tests cannot differentiate which efferent system component is dysfunctional, so management strategies should be generalised to improving overall efferent function (Table 3).

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TABLE 3. Tests of efferent function

Efferent function	Clinical test	Uses	Equipment needed	Facilitators	Barriers
Strength assessment	Isokinetic dynamometry	Assess concentric or eccentric strength of quadriceps and hamstrings through a specified range of motion	Isokinetic dynamometer	Reliable measurement. Able to assess strength through a specified range of motion.	High cost
	Repetition maximum testing	Assess an athlete's ability to perform a movement under increasing load to determine a 1,3 or 5 repetition maximum	Gym equipment	Test can be independently completed by the athlete. Can test several movement patterns.	Can take time to complete the test when allowing for appropriate warm up and rest in between sets. If using a compound exercise for the testing protocol, then this may allow for compensatory movements to hide weakness in a specific muscle.
	Isometric strength testing	Able to assess isometric strength by fixing a handheld dynamometer with non-elastic bands to an anchored limb	Handheld dynamometer and non-elastic bands	Provides reliable measure of isometric strength	Can miss range-specific strength asymmetries

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Voluntary activation	Central activation ratio	Assess the athlete's muscle recruitment. Maximum voluntary contraction is completed and output is measured. A superimposed electrical burst is then used to contract the muscle to get a maximum output. A percentage score can then be derived based off how well athlete can contract muscle in relation to maximum output available	Isokinetic dynamometer and electrical stimulation device	A reliable measure of how well an individual can recruit a muscle	Cost of equipment is high and expertise is needed to conduct the test
	Surface EMG	Assess how well the athlete can recruit a muscle in comparison to the contralateral muscle during a maximum voluntary contraction	Surface EMG device	Easy to use and more accessible for individuals	Cost of equipment can be high.
Motor cortical activity	Functional brain MRI	Assess if altered processing is occurring in the motor cortex during a movement task (eg. Knee flexion and extension)	MRI machine and specialist software	Identifies areas of brain active during movement	Financial and time cost. Expertise to interpret the scan.
	Electroencephalogram	Assess if altered processing is occurring in motor cortex during a movement task (eg. Knee flexion and extension)	EEG device and specialist software	Identifies areas of brain active during movement	Financial and time cost Expertise to interpret the scan
	Transcranial magnetic stimulation	Assess the excitability of the motor cortex via the application of magnetic field pulses which induce electrical currents in the brain	Transcranial magnetic stimulation device, cortical mapping software, EMG and sometimes EEG	Identifies excitability of motor cortex	Financial and time cost. Expertise to interpret the results

Descending motor pathway	Transcranial magnetic stimulation	Assess the excitability of the descending motor pathway via the application of pulses to the motor neurones and measuring motor-evoked potentials in the target muscles	Transcranial magnetic stimulation device, cortical mapping software, EMG and sometimes EEG	Identifies excitability of descending motor pathway	Financial and time cost. Expertise to interpret the results
Spinal reflex	Hoffmans reflex	Assess the excitability of the spinal reflexive pathway	EMG, isokinetic dynamometer and electrical muscle stimulator	Accurate assessment of spinal reflexive excitability	Financial and time cost. Expertise to interpret the results

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290 Quadriceps and hamstrings strength can be tested via repetition maximum testing (1, 3 or 5 repetition
291 maximum), handheld dynamometry or isokinetic dynamometry. Although they do not provide
292 equivalent measures,[18] repetition maximum testing is best placed to examine strength changes
293 through a specific movement such as squats, knee extensions or deadlifts[18] which may help clinicians
294 understand if the athlete's functional patterns are improving. A limitation of using repetition
295 maximum strength testing is that multi-joint movements may improve due to improved inter-
296 segmental coordination, which may mask any lack of improvement in isolated quadriceps strength.

297 Isokinetic dynamometry is the gold standard for examining muscle strength (Appendix D).[15,18] It has
298 high reliability (ICC= 0.74 – 0.93)[30,91] when assessing quadriceps and hamstrings peak torque. Elite
299 male field-based athletes should be achieving a quadriceps and hamstrings peak torque of 3.19±0.1
300 Nm/kg and 1.78±0.1 Nm/kg; elite female field-based athletes should achieve values of 2.32±0.36
301 Nm/kg and 1.36±0.21Nm/kg.[101] In our practice, athletes falling below these levels are prescribed
302 strategies to improve strength and efferent function. Although access to isokinetic dynamometry is
303 improving, it may still be out of reach for many clinicians. Handheld dynamometry is a cost-effective
304 alternative. It is reliable and valid in measuring isometric quadriceps strength when using non-elastic
305 straps.[1,46] One disadvantage is that it cannot identify range-specific strength deficits.[32] Strength
306 assessments should be completed every 6-8 weeks.

307 Deficits in quadriceps voluntary activation persist following ACLR.[48] Surface EMG biofeedback devices
308 are accessible and can determine the magnitude of motor unit recruitment during a contraction,
309 which can give insight into any potential altered efferent function. Commercially available surface
310 EMG biofeedback units (e.g. 'mTrigger', LLC, *Delaware, USA,* and Neurotrac Myo Plus 2, *Verity Medical*
311 *Ltd*) have reported that they can compare EMG activity between limbs via dual-channel EMG. The
312 athlete performs 3-6 repetitions of maximal isometric quadriceps activation on the injured and
313 healthy limbs, with the surface electrodes placed over the motor points of the quadriceps. Differences
314 of more than 22%-30% may be clinically-relevant[10,67] and can guide clinicians to implement

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appropriate strategies to improve efferent function.[3,92] We suggest assessing every 2 weeks until no significant difference is noted (Appendix E).

Summary

Sensorimotor dysfunction may persist following ACL injury and occurs across all aspects of the sensorimotor system: afferent pathways, efferent pathways and central processing (Figure 2). We recommend clinicians focus on assessing the somatosensory, visual and efferent systems. Routine assessment of the somatosensory system (JPS testing, pain and effusion testing) and visual system (visual-motor reliance testing and visual-motor processing testing) will guide clinical decisions. Assessing efferent function via strength testing and EMG assessment for voluntary activation can be easily implemented into clinical practice.

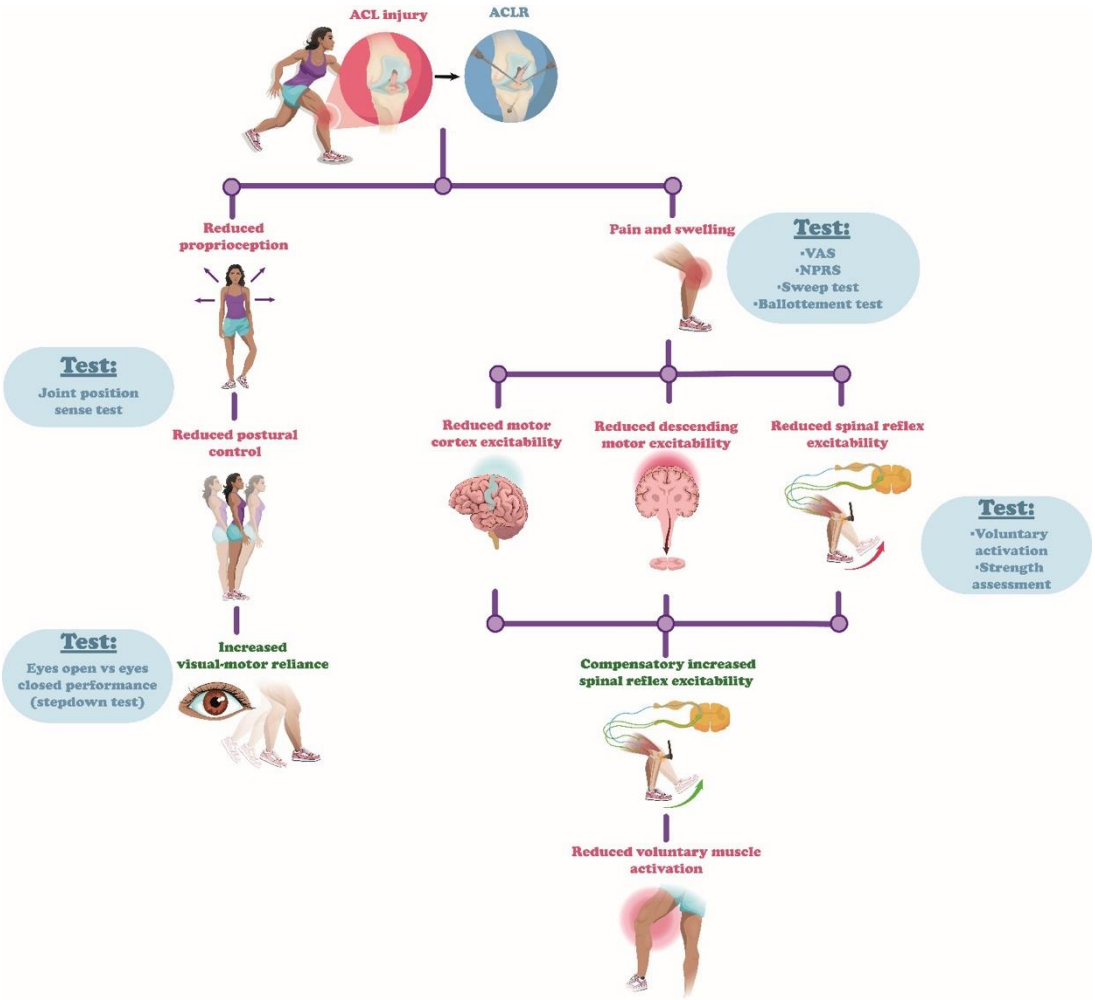

Figure 2. Stages of sensorimotor dysfunction occurring following ACL injury and ACLR

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Key Points

Findings: Sensorimotor dysfunction can occur following an ACL injury. It occurs across the somatosensory, visual and efferent systems.

Implications: All three systems should be assessed: somatosensory (JPS, pain and effusion tests), visual system (visual-motor reliance and visual-motor processing tests), and efferent system (strength and voluntary activation tests)

Caution: More research is required for visual-motor reliance and visual-motor processing tests to identify normative values, and to determine if they are related to reinjury

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Appendix A

Joint position assessment

Relph and colleagues[79] were able to demonstrate via an image recorded angulation test that elite athletes, even a year post-ACLR, had reduced JPS function as compared to sport and age-matched athletes.[79] To complete the testing method as per the study,[78,79] have the athlete sitting with their knee over the edge of a plinth. The athlete is blinded throughout the test. Then, place three markers placed on their leg: (1) lateral femoral condyle, (2) lateral malleolus, and (3) a point along a line between the greater trochanter and lateral femoral condyle (see Figure A1). Setup a camera on a tripod 3 metres away in-line with the lateral femoral condyle. Passively extend the athlete's knee (with hand placement under the distal shin) to the target angle (which was randomly selected within a range of 30-60°). Hold the knee at the target angle for 5 seconds. Passively bring the athlete's knee back to the starting position (leg comfortably relaxed in naturally flexed position whilst sitting). Then, instruct the athlete to actively extend their knee to the angle that they believe it was held at by the clinician. A photograph is taken on 6 occasions: first at the target angle when the leg was passively held there, and then for each of the 5 repetitions when the participant stops actively moving their knee and reports that they are at the target angle. The angles are then measured from the photographs using digitising software, such as 'autoCAD' (*Autodesk, USA*) or free software such as Kinovea (https://www.kinovea.org) and ImageJ (https://imagej.net/ij), and the difference between the target angle and the athlete's attempted replication of the angle is calculated. The average of difference between each of the 5 reps is taken as the measurement of JPS ability. The testing procedure has good inter-rater reliability (ICC=0.98) and intra-rater reliability (ICC=0.96)[78], with a smallest detectable change of 1.10° for knee flexion (standard error of measurement 0.40°) and 1.35° for knee extension (standard error of measurement 0.60°) [78]. A difference of greater than 5.3° between the ACLR limb and the uninjured limb of the athlete represents poor proprioceptive ability in the ACLR limb [79].

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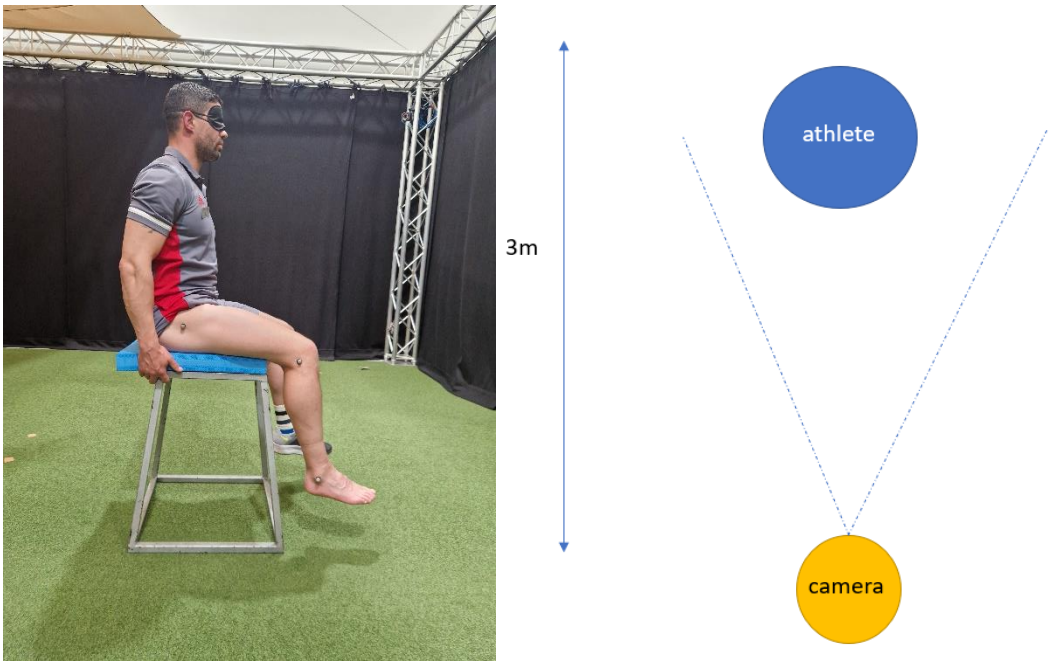

FIGURE A1. Joint position assessment set up

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Appendix B

Effusion assessment

Supplementary video on Effusion tests

(insert video)

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Appendix C

Stepdown test for assessing visual-motor reliance

The following protocol is currently unpublished. A force plate is required for this assessment. A 20-centimetre step is placed behind the force plate. The athlete is barefoot for the test. The athlete starts on the step with their hands on their hips. They are instructed to step onto the force plate with a single leg and achieve stability as quickly as possible whilst also maintaining their landing posture for 20 seconds. The landing posture of the leg is not standardised. Three trials are conducted for both legs. The athlete then repeats the procedure whilst wearing a blindfold to obstruct vision. Trials were deemed invalid if the non-testing leg touches the ground, a double contact occurred with the testing foot, the blindfold was removed or their hands were removed from their waist.

The medial-lateral and anterior-posterior data are then processed using a sequential estimation technique[5,9] to calculate time to stability (TTS) in each direction (medial-lateral and anterior-posterior). A visual-motor reliance index is then calculated using the TTS eyes open versus the eyes closed:

$$VR\ index = \left(\frac{[TTS\ (blinded) - TTS\ (full\ vision)]}{TTS\ (full\ vision)}\right)\ x\ 100$$

(see stepdown Video)

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Appendix D

Isokinetic dynamometer assessment of quadriceps and hamstrings strength

A systematic review[100] recommended a standardised protocol when assessing quadriceps and hamstrings strength with an isokinetic dynamometer. It is the current protocol used at the Sports Surgery Clinic (Dublin, Ireland) as part of its ACL research program.[62,63] The participant is seated comfortably with the torso and limb to be tested fixed to the chair and testing arm, respectively. Three sets of five repetitions (first set being a sub-maximal familiarisation set) of concentric knee flexion and extension are completed. A minute rest is provided between each of the sets and the knee is tested between a range of 0-100°, at a speed of 60°/second, and gravity correction employed. Peak torque of the quadriceps and hamstrings represent their strength. Values that athletes should aim to return to are quadriceps peak torque of 240% to 270% body mass, and hamstrings peak torque of 150% to 160% body mass.[63]

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Appendix E

Use of surface EMG biofeedback devices for assessment of voluntary quadriceps activation

Two electrodes are placed over the vastus medialis and two over vastus lateralis (see Figure 2). Athletes sit upright in a chair with their back supported, extend their knee to neutral and contract their quadriceps maximally. Several repetitions are performed until a consistent EMG recording is found. The athlete then lies on their back with their leg extended. They contract their quadriceps maximally and perform a straight leg raise. Several repetitions of the straight leg raise are performed until a consistent EMG recording is taken. A reduction in voluntary quadriceps activation is determined by a reduction in the EMG recording in the ACLR limb as compared to the uninjured limb either during the straight leg raise or the knee extension task. Due to the potential for noise created by electrode placement being slightly different, we suggest that a difference of greater than or equal to 20-30% is what is used in the clinic to determine if there is a reduction in voluntary activation.[10,67]

FIGURE A2. Electrode placement for surface EMG assessment of the quadriceps.

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